

KAVYA SREE CHANDHI

Software Engineer | Python | C++ | AWS | SQL | Generative AI

Mail id: chandhikavyasree@gmail.com | Contact Number: +1 9406298898 | Location: Herriman, Utah, 84096

LinkedIn: <https://www.linkedin.com/in/chandhi-kavya-sree/> |

GitHub: <https://github.com/kavya-sree-chandhi> | Portfolio: <https://kavya-sree-chandhi.github.io/> |

PROFESSIONAL SUMMARY

Software Engineer with **5+ years** of experience designing and developing scalable backend applications, automation solutions, and AI-powered systems using Python, C++, SQL, and AWS. Strong foundation in data structures, algorithms, object-oriented programming, and system design. Experienced in building REST APIs, data processing pipelines, and cloud-based applications across healthcare, education, finance, and automotive domains. Skilled in performance optimization, debugging, CI/CD, and deploying reliable software solutions in Agile environments.

TECHNICAL SKILLS:

- **Programming Languages:** C, C++, Python, JavaScript, SQL
- **Core Computer Science:** Data Structures & Algorithms, Object-Oriented Programming, Memory Management, Multithreading, Operating Systems, System Design
- **Software Development:** STL (Standard Template Library), Pointers, File Handling, Debugging, Profiling, Performance Optimization
- **Frameworks & Libraries:** FastAPI, Flask, OpenCV, Scikit-learn
- **Generative AI:** LLMs, RAG, Prompt Engineering, Fine-Tuning, AI Agents, LangChain, LangGraph, Hugging Face, OpenAI API, Ollama, DALL-E, Vector Databases (Milvus, ChromaDB, Pinecone)
- **Machine Learning & Data Science:** Supervised and Unsupervised Learning, Predictive Modeling, Feature Engineering, Cross-Validation, Hyperparameter Tuning, Statistical Analysis
- **Deep Learning & Computer Vision:** ANN, CNN, RNN, LSTM, ResNet, VGG, MobileNet, Transfer Learning, GANs, Autoencoders, PyTorch, TensorFlow, Keras
- **Natural Language Processing:** Sentiment Analysis, Summarization, Named Entity Recognition (NER), Emotion Detection, Text Classification, spaCy, NLTK
- **Recommender Systems & Optimization:** Collaborative Filtering, Matrix Factorization, Bayesian Optimization, Optuna, Hyperopt
- **Big Data & Streaming:** Apache Spark, PySpark, Hadoop, Kafka, Spark Streaming, Snowflake, Redshift, Azure Data Lake, BigQuery
- **MLOps & DevOps:** Docker, Kubernetes, Jenkins, GitHub Actions, MLflow, Weights & Biases (W&B), DVC, Terraform, Airflow, Helm, CI/CD
- **Visualization & Analytics:** Power BI, Tableau, Grafana, Matplotlib, Plotly, Dash, Excel, Exploratory Data Analysis (EDA)
- **Cloud Platforms:** AWS (SageMaker, Lambda, EC2, S3, RDS, Glue), Azure (Prompt Flow, Bot Framework, Cosmos DB, Data Factory)
- **Databases:** MySQL, PostgreSQL, MongoDB, DynamoDB, Neo4j, NoSQL
- **Collaboration & Tools:** Git, Jira, Confluence, Jupyter Notebook, VS Code, PyCharm, Anaconda

WORK EXPERIENCE:

Global IT Link Solutions (Contract)

Aug 2024 – Jan 2026

Client - Intermountain Health

Role - Software Engineer

Project Description: Worked as a contract Software Engineer through Global IT Link Solutions for Intermountain Health, developing backend applications, automation solutions, and cloud-based services to support healthcare operations and improve system efficiency, reliability, and performance.

Key Responsibilities:

- Developed scalable backend applications and automation solutions using Python, C++, SQL, and AWS.
- Designed and implemented REST APIs and data processing workflows to support business-critical healthcare applications.

- Optimized application performance resolved production issues, and enhanced overall system reliability.
- Implemented CI/CD pipelines using GitHub Actions, Docker, and AWS to automate build, testing, and deployment processes.
- Developed SQL queries and scripts for data extraction, transformation, and reporting.
- Collaborated with cross-functional teams in Agile/Scrum environments to deliver high-quality software solutions.
- Participated in code reviews, debugging, and performance tuning to ensure maintainable and efficient code.
- Monitored application health and supported production releases and issue resolution.
- Created technical documentation and contributed to knowledge sharing across the team.

Client – DXC Technology, India

Aug 2022 – Aug 2023

Role – Associate Professional Software Engineer

Project Description: Enhanced Product Lifecycle Management (PLM) systems for global automotive engineering teams by developing Python automation tools, backend modules, reporting pipelines, and predictive analytics solutions to improve workflow efficiency, system reliability, and operational visibility.

Key Responsibilities:

- Developed Python automation scripts to streamline Product Lifecycle Management (PLM) workflows, reducing manual effort by 30%.
- Designed and implemented user management modules supporting 5,000+ engineering accounts with secure role-based access.
- Built backend components and data processing logic to support engineering workflows and system integrations.
- Optimized SQL queries and reporting pipelines, accelerating data extraction and operational reporting.
- Integrated machine learning models for predictive maintenance, reducing system downtime by 15%.
- Created interactive dashboards using Tableau and Grafana to monitor license utilization, system performance, and usage trends.
- Developed CI/CD pipelines using Docker and Jenkins to automate deployment of PLM automation solutions.
- Authored technical documentation and standard operating procedures to support onboarding and knowledge transfer.
- Collaborated with engineering and IT teams to deliver data-driven insights and ensure compliance with ISO/IEC standards.

Client – Pelican IT Solutions Pvt. Ltd., India

Feb 2020 – July 2022

Role – Data Analyst

Project Description: Supported the development of a Fraud Detection and Risk Analytics Platform for a leading banking client by building Python-based data processing pipelines, anomaly detection models, ETL workflows, and real-time dashboards to identify suspicious transactions and strengthen regulatory compliance.

Key Responsibilities:

- Developed Python and SQL data processing pipelines to analyze large-scale financial transaction data.
- Built anomaly detection and machine learning models, including Random Forest, Isolation Forest, and Autoencoders, improving fraud detection accuracy by 20%.
- Automated ETL workflows using Python (Pandas, NumPy, Scikit-learn), reducing manual effort by 30%.
- Designed real-time fraud monitoring dashboards using Flask and Grafana.
- Developed fraud risk indicators such as transaction velocity, behavioral patterns, and seasonal trends.
- Optimized data preparation and scoring workflows to improve near real-time fraud analysis performance.
- Collaborated with compliance and risk teams to ensure alignment with KYC/AML regulations.

VOLUNTEERING:

Client – Network Theory Applied Research Institute

Jun 2025 – Aug 2025

Role – Software Engineer (Volunteer)

Project Description: Contributed to the development and enhancement of NTARI's internal research operations platform, focusing on frontend development and backend system analysis within an Agile team environment.

Key Responsibilities:

experience.

- Contributed to frontend architecture implementation, ensuring modular and maintainable code structure.
- Conducted detailed backend file and data structure analysis to understand system logic and integration points.
- Collaborated with cross-functional teams in an Agile environment to deliver incremental feature updates.
- Assisted in aligning frontend components with backend workflows to ensure accurate data representation and system consistency.
- Participated in sprint discussions, issue tracking, and iterative development cycles to support timely feature releases.
- Identified opportunities for performance improvements and workflow optimization across system components.

Client –Street Cause SWEC (NGO Initiative)

July 2020 – July 2021

Role – Associate President – Executive Board (2020–2021)

Project Description: Led social impact initiatives focused on community welfare during the COVID-19 pandemic, coordinating large-scale relief drives including slum adoption programs and public health awareness campaigns. Managed cross-functional volunteer teams to execute essential supply distribution, mask drives, and community outreach programs impacting underserved populations.

Key Responsibilities:

- Co-led strategic planning and execution of Slum Adoption Project, supporting 10+ slums and 262+ families with monthly essential groceries.
- Organized and managed COVID-19 Mask Distribution Drive (June 2020), ensuring community safety awareness and protective supply distribution.
- Supervised multiple functional departments including Women Empowerment, Education, and Orphanage & Old Age Homes initiatives.
- Coordinated volunteer teams, task delegation, and execution timelines to ensure smooth program delivery.
- Managed stakeholder communication, partnerships, and operational planning.
- Contributed to digital campaign promotion and awareness initiatives through social media outreach.
- Strengthened organizational governance and operational efficiency as part of the Executive Board.

ACADEMIC PROJECTS:

AI Research Agent for Healthcare Diagnostics

- Built an AI-powered research assistant using Python, Streamlit, LangChain, and LLaMA3 with a RAG pipeline (FAISS + PyMuPDF), incorporating citation validation to reduce hallucinations and generate structured healthcare diagnostics reports.

Image Classification for Disease Detection

- Designed deep learning models (CNNs, VGG16, ResNet50) for MRI and X-ray scans, achieving 98% accuracy in brain tumor detection and improving robustness with transfer learning.

Youth Smoking and Drug Use Analysis

- Conducted EDA and ML modeling on 8,993 records using Python, revealing peer influence and school-based interventions as key factors in reducing substance use.

Securing Data Using Blockchain and Artificial Intelligence

- Built SecNet architecture with Blockchain + AI anomaly detection on a Django prototype, enabling patient-controlled healthcare data access and tamper-proof record management.

CERTIFICATIONS:

- **Microsoft Azure Fundamentals (AZ - 900) - Microsoft**
- **Artificial Intelligence Fundamentals - IBM**
- **Google Technical Support Fundamentals – Coursera/ Google**

ACADEMICS:

Master of Science in Computer Science | GPA: 3.916/4

Aug 2023 – May 2025

University of North Texas, Denton, Texas